

Sample End Semester Question Paper - 2025

Subject: Strength of Materials | Code: CE301 | Semester: 3

Branch: Civil | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define slenderness ratio.
2. What is a point of contraflexure?
3. Name one theory of failure.
4. Define shear stress.
5. What is a built-up section?
6. Define stress and strain.
7. State Hooke's law.
8. What is a bending moment diagram?
9. Define section modulus.
10. What is a column?

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain stress and strain. Derive the relation $E = 2G(1 + \nu)$.
2. Draw SFD and BMD for a cantilever with UDL.
3. Explain shear stress distribution in a rectangular section.
4. Explain shortening of a column under eccentric load.
5. Explain slenderness ratio in RCC columns.
6. Solve a built-up section problem for moment of inertia.